



Assignment 1 – Q2, Q4, Q6

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Outline

1. Q2 – Object Tracking
2. Q4 – Classification
3. Q6 – Autoencoder

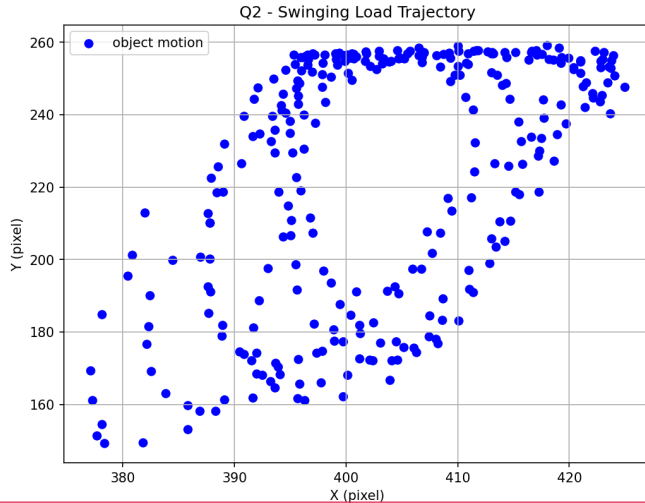
Q2 – Object Tracking

5 points

Objective & Method

- **Objective:** track the swinging load in `test_swingingload.avi` and show its trajectory as it swings
- **Method:**
 - Detector: the YOLO model trained in `Q1 (best.pt)`
 - `model.track(frame, persist=True)` on every frame – keeps the same object identity across frames
 - Bounding-box center (c_x, c_y) extracted for every frame the load is detected in
 - Plot all centers as a scatter of pixel coordinates X vs. Y

Result



Q4 – Classification

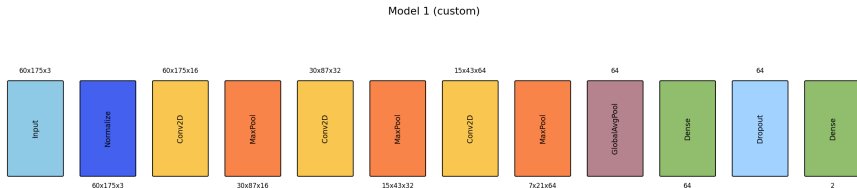
10 points

Objective & Data

- **Objective:** determine motor state of health (healthy vs. faulty) and compare an own CNN against the baseline architecture given in class
- **Data:** fault-classification-class – vibration scalograms (laser vibrometer $\rightarrow$ wavelet transform), $60 \times 175 \times 3$ px
- 180 measurements per class, each split into 5 time-window crops $\Rightarrow$ 900 images/class
- Fixed 80/20 split by measurement number (1–144 train, 145–180 val) – all crops of one measurement stay together, no data leakage

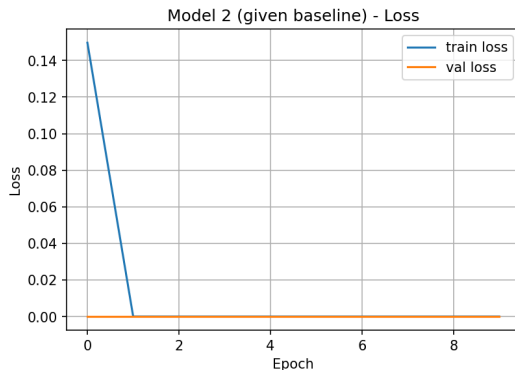
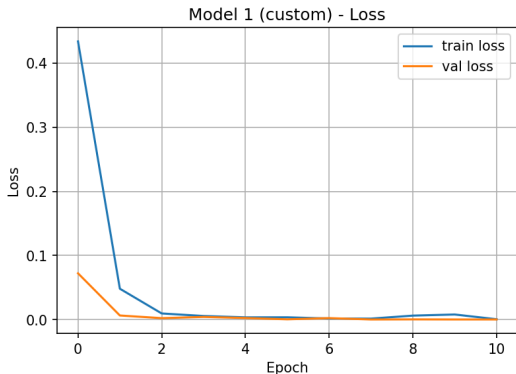
	Training	Validation	Test
Images	1440	360	200

Model 1 (own) vs. Model 2 (given baseline)



- **Model 1 (custom, 27.9k params):** $3 \times [\text{Conv2D} + \text{MaxPool}]$ ($16 \rightarrow 32 \rightarrow 64$), GlobalAveragePooling, Dense(64)
- **Model 2 (given, 105k params):** $5 \times [\text{Conv2D} + \text{MaxPool} + \text{ReLU}]$ ($8 \rightarrow 16 \rightarrow 64 \rightarrow 64 \rightarrow 64$), Flatten(320), FC(64)

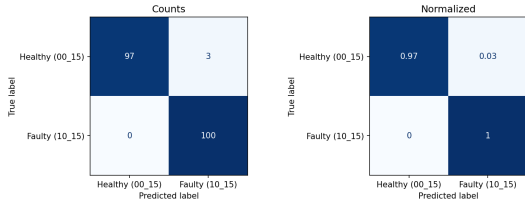
Training



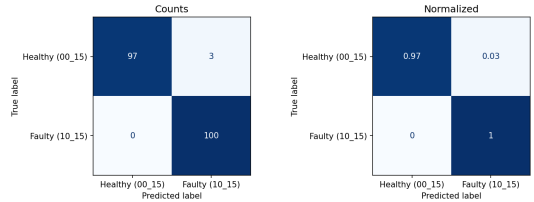
Adam, categorical cross-entropy, EarlyStopping on validation loss (patience 3, best weights restored)

Evaluation

Model 1 (custom)



Model 2 (given baseline)



Comparison & Conclusion

Model	Params	Inference	Accuracy	Precision	F1
Model 1 (custom)	27,874	0.76 ms	0.985	0.985	0.985
Model 2 (baseline)	105,202	0.76 ms	0.985	0.985	0.985

- Both models are close to perfect – the two classes are visually very distinct in the scalograms
- Model 1 reaches the **same accuracy with $3.8\times$ fewer parameters**, mainly from GlobalAveragePooling instead of Flatten

Q6 – Autoencoder

10 points

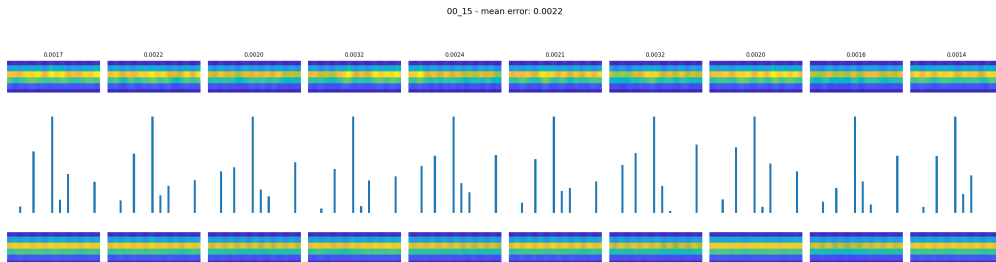
Objective & Method

- **Objective:** compress each scalogram to a latent vector and reconstruct it, for 10 random images per class
- **Method:** dense autoencoder (matches the architecture shown in class)

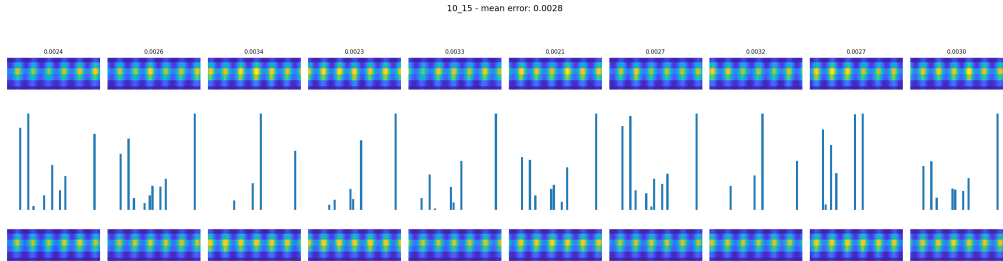


- Trained unsupervised: target = input itself, loss = MSE (reconstruction error)

Result – Healthy (00_15)



Result – Faulty (10_15)



Conclusion

Class	Mean reconstruction error (MSE)
Healthy (00_15)	0.0022
Faulty (10_15)	0.0028

- Both classes reconstruct well – the 32-value latent vector preserves the overall band pattern
- Faulty images have a slightly higher error: the periodic peak pattern carries more high-frequency detail than the smooth healthy band, making it harder to compress into 32 numbers